MIL-I-13145(Ord)
4 December 1953
SUPERSEDING
AXS-1723
12 March 1946
PA-PD-243
16 June 1953

MILITARY SPECIFICATION

IGNITER, ELECTRIC, T14E2
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers loading, assembling and packing for the T14E2 Electric Igniter for use in 4.5 inch rockets.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.3 The following specifications, drawings, and publications, and all drawings and specifications referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-P-223 -Powder, Black
MIL-G-2550 -General Specification for Ammunition Except Small Arms Ammunition

DRAWINGS

ORDANCE CORPS

82-5-95 -Igniter, Electric, T14E3. Assembly and Details

PUBLICATIONS

ORDANCE CORPS

ORD-STD-M16-180-Igniter, Electric, T14E2. Loaded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:07Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. -Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 The igniter shall comply with all requirements specified on Drawing 82-5-95, and all drawings referenced thereon, and with all requirements specified in applicable specifications.

3.3 Black Powder. -The moisture content of the black powder after loading, shall not exceed .7 percent when tested as specified in 4.4.

3.4 Functioning. -The igniters shall not misfire and shall function with an ignition delay not exceeding .10 second when tested as specified in 4.6.

Note: A misfire shall be defined as a failure of the igniter to function due to defects within the igniter.

Note: Ignition delay shall be defined as the time elapsed form initiation of the current to initiation of burning of black powder.

3.5 Workmanship. -All parts shall be free of chips, dirt, grease and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot. -A lot shall consist of all igniters loaded by one manufacturer in one unchanged process in accordance with the same drawing, same drawing revision, same specification and smane specification revision. Each lot shall contain:

a. squibs from not more than one lot, and
b. black powder from not more than one lot.

4.2 Sampling. -Number of samples, acceptance and rejection criteria used for determination of lot acceptance for test as specified herein shall be in accordance with Standard Inspection Procedure ORD-SIP-M16-180. Sampling of component material is specified in paragraphs covering test of materials.

4.3 Inspection. -Inspection shall be as specified is Specification MIL-G-2550.

4.4 Determination of moisture content of black powder. -The moisture content of black powder shall be determined at intervals as directed by the inspector. At least one moisture determination per each lot of igniters shall be made on black powder taken from complete assemblies to determine compliance with 3.3. The method of determining the percentage of moisture shall be as specified in Specification JAN-P-223.

4.5 Vacumm Test. -Sample igniter assemblies shall be stored for one hour at 160° ± 5°F, and then for one hour at -65° ± 5°F. The time delay between the conditioning stages snail not exceed 5 minutes. Within one hour after removal from the conditioning chamber, the igniters shall be subjected to an external vacuum as specified on the drawing using the apparatus shown on figure 1. Observe for leakage as indicated by the formation of any air bubbles during the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:07Z
Check the source to verify that this is the current version before use.

4.6 Functioning. -Additional sample igniters shall be conditioned in accordance with the procedure specified for the vacuum test (see 4.5) and then shall be completely submerged (including lead wires) under a depth of 11 to 12 inches of water at 70° 10°F for a period of 4 hours minimum. Each sample be proof-fired in a circuit (see figure 2) within a minutes after removal from the water to determine compliance with 3.4. The igniters shall be proof-fired confined in a steel tube 14 ± 1/2" long by .9 /.l'' inside diameter by 1/16" minimum wall thickness. Prior to connection of the igniter assemblies to the test circuit, the circuit shall be prepared for firing by measuring the resistance of the connection wires from switch "S" (see figure 2) to the igniter assembly. The resistor "r" shall then be set to 1.00 ~ .05 ohm plus the measured connector wire resistance. Switch "S" shall be closed on resistor "r" and the resistor "R" so adjusted that milliapmeter (MA) reads 545 ± 5 milliamperes. Switch "S" shall be opened several seconds and then closed again; the current shall reproduce within ± 5 milliamperes. Following preparation of the test circuit, the assemblies shall be fired using the circuit shown in figure 2, behind a suitable barricade. Before every third or fourth firing, or before the first firing after a period of inactivity, the current shall be checked by closing switch "S" on resistor "r" and readjusting resistor "R" if necessary. The battery shall be discarded if the current becomes highly unstable, or if its voltage, measured with a voltmeter whose sensitivity is 110 ohms per volt or better, is less than 2.8 volts when no other current is being drawn from the battery.

5. PREPARATION FOR DELIVERY

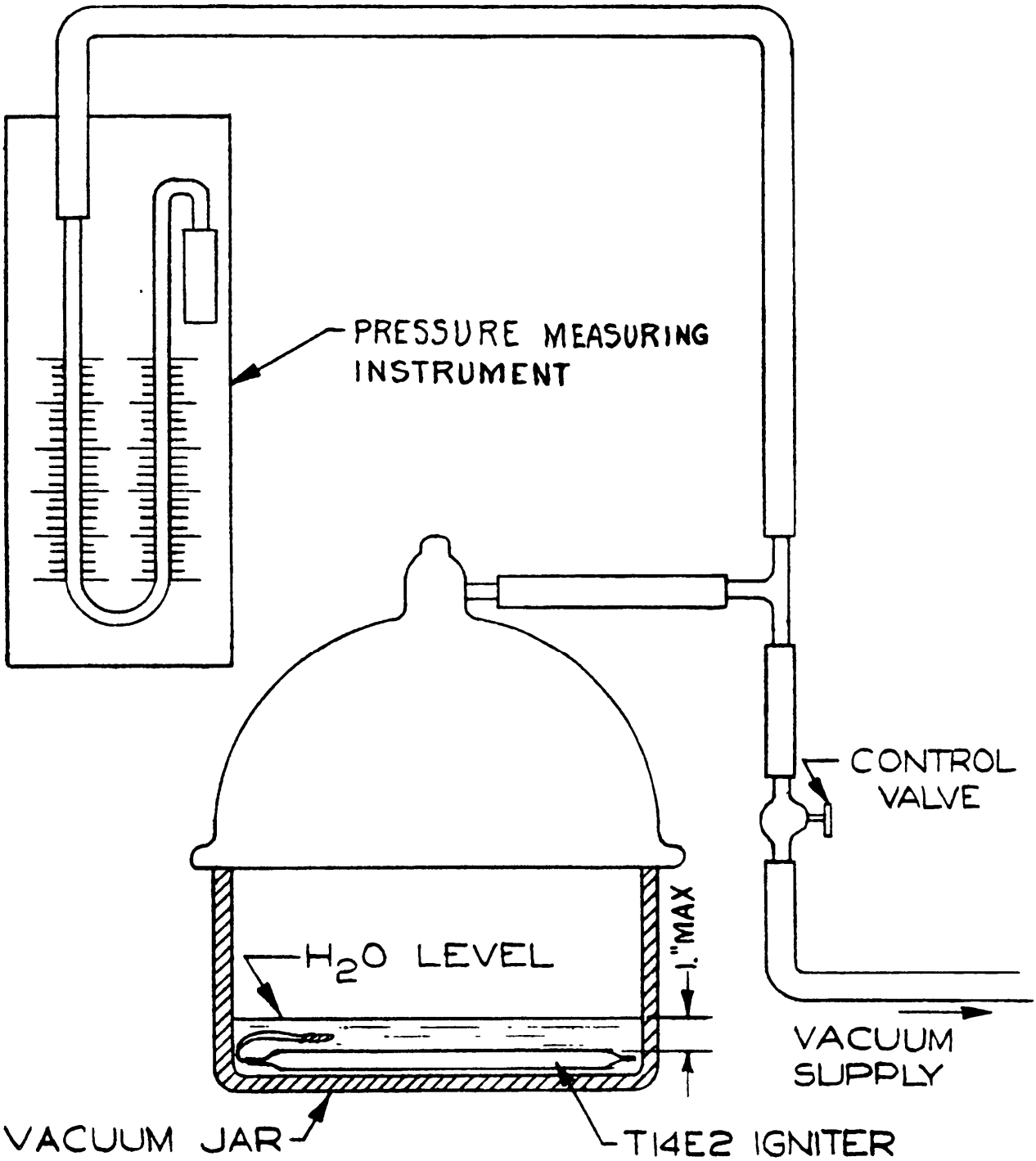
5.1 Packaging, packing, labeling and marking.-Packaging, packing, labeling and marking shall be as specified on the applicable drawing.

5.2 Data cards. -Data card information shall be as specified in Specification MIL-G-2550.

NOTICE: This specification, together with specification and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

Custodian:
Army-Ordnance Corps

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:07Z
Check the source to verify that this is the current version before use.



APPARATUS FOR VACUUM TEST OF IGNITER, ELECTRIC T14E2

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:07Z
Check the source to verify that this is the current version before use.

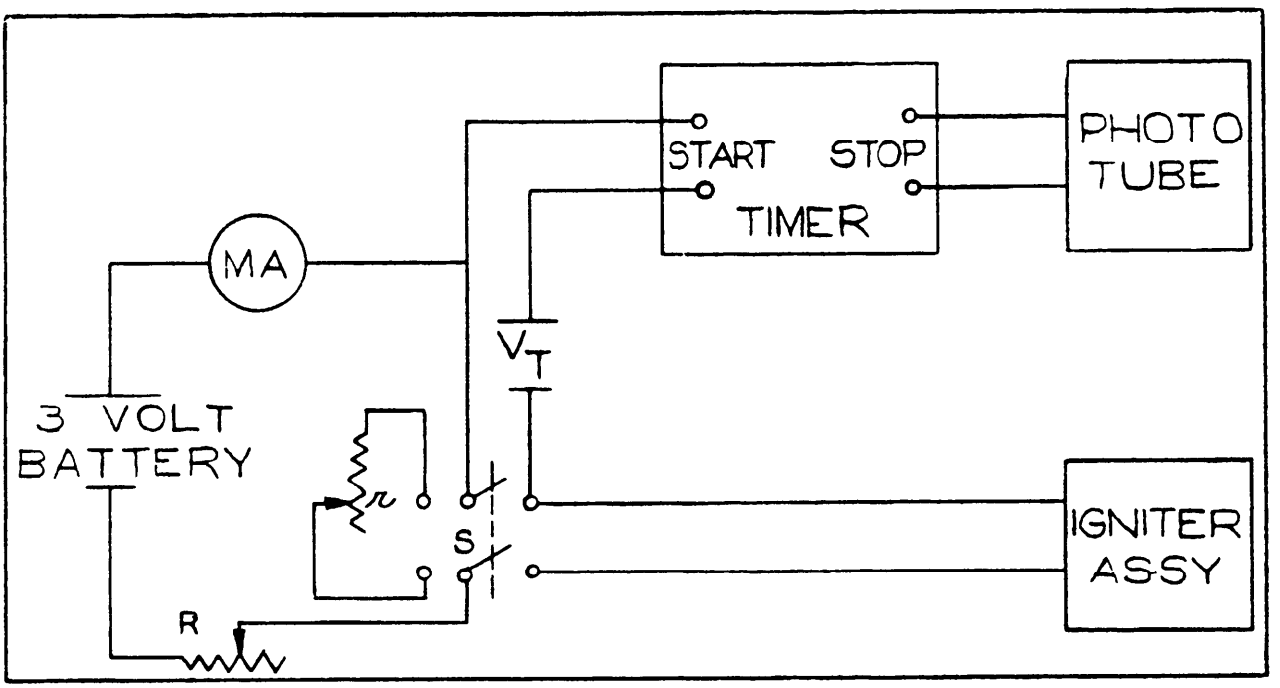


PHOTO TUBE SHOULD HAVE S-1 SPECTRAL RESPONSE.
V_T = VOLTAGE SUPPLY TO START TIMER AS REQUIRED.

FIRING INSTRUCTIONS:

1 - OPEN SWITCH "S" ATTACH IGNITER ASSEMBLY, AND SET TIMER.

2 - PLACE SWITCH "S" IN FIRING POSITION (TO RIGHT IN DIAGRAM) AND RECORD FUNCTIONING TIME AS GIVEN BY TIMER.

FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:07Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:07Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 202610-18T00:07Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING OFFICER
PICATINNY ARSENAL
ATTN: SMUPA-DC7
DOVER, NEW JERSEY

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:07Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:07Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:07Z
Check the source to verify that this is the current version before use.

NOTICE OF INACTIVATION
FOR NEW DESIGN

INCH POUND

MIL-I-13145(AR)
NOTICE 2
22 April 1996

MILITARY SPECIFICATION

IGNITER, ELECTRIC, T14E2 LOADING, ASSEMBLING AND PACKING

This notice should be filed in front of MIL-I-13145(AR), dated 4 December 1953

MIL-I-l3145(AR), dated 4 December 1953. with Validation Notice 1, dated 3 October 1991 is inactive for new design an is no longer used, except for replacement purposes.

Preparing Activity:
Army-AR

AMSC N/A FSC 1340

DISTRIBUTION STATEMENT A. Approved for public release. distribution is limited.

Source: https://assist.dla.mil -- Downloaded: 2026l0-18T00:07Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:07Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:07Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T00:07Z
Check the source to verify that this is the current version before use.